



Emotion-semantic interaction network for fake news detection: Perspectives on question and non-question comment semantics

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ABSTRACT

Current fake news detectors often overlook the association between news content emotion and comment semantics, especially questioning vs. non-questioning language. We observe that fake news evokes distinct comment patterns: non-sad emotions (e.g., anger, surprise) in fake content drive more questioning semantics (e.g., “Where is the video to support this?”) and suppress non-questioning replies, while sadness shows the opposite trend. Ignoring this emotion-semantic interaction limits detection accuracy. To address this limitation, we propose an Emotion-Semantic Interaction Network (ESIN), which learns a latent association between content emotions and comment semantics from questioning and non-questioning perspectives. Specifically, ESIN incorporates the interaction of content emotion with original comment semantics, as well as the distribution of extracted comment semantic categories (i.e., question and non-question), achieved through our proposed mechanisms called Codebook Initialization and Semantic Quantification, based on cross-attention. The ESIN model is comprehensively evaluated on two widely used fake news datasets, namely RumourEval and WeiBo. The ESIN achieves competitive performance, outperforming baseline models by a significant margin of 5.27% and 4.08% in weighted F1-score (F1.) and accuracy (Acc.) on the RumourEval dataset, and by 2.43% and 2.38% in F1. and Acc. on the WeiBo dataset. The promising result verifies the effectiveness of our ESIN model. Theoretically, this study fills a critical gap by highlighting how content emotions shape comment semantics (questioning vs. non-questioning) as a veracity cue, advancing understanding of user engagement patterns in fake news spread. Practically, ESIN offers actionable strategies for platforms to flag high-risk content and equips policymakers with tools for user training and evidence-based regulations, enhancing misinformation mitigation and public trust in digital information ecosystems.

1. Introduction

The rapid expansion of social media platforms in recent years has revolutionized how news is accessed, disseminated, and discussed. Social networking sites enable real-time sharing and commentary on news content (Mitchell et al., 2016), which has democratized information flow and facilitated global discourse. However, this accessibility has also brought about a significant challenge: the widespread circulation of fake news. Fake news undermines public trust and disrupts societal stability in areas such as politics (Fisher et al., 2016) and economics (ElBoghdady, 2013). As a result, the detection of fake news has become a critical task for safeguarding the integrity of information.

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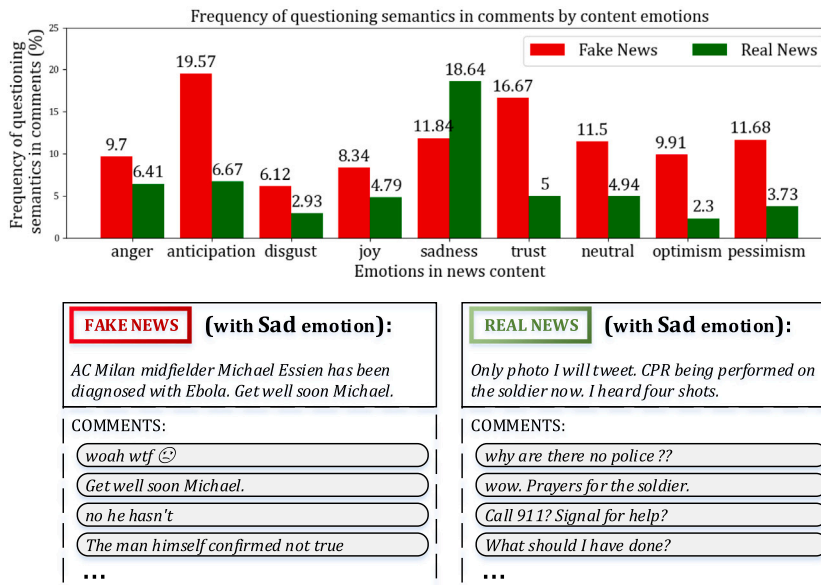


Fig. 1. The frequency of questioning semantics in comments by content emotions observed in a fake news dataset (the combination of RumourEval Gorrell et al., 2019 and WeiBo Ma et al., 2016), which is distinguishable in fake versus real news, and two cases from the dataset.

In the early stages of fake news detection research, knowledge-based methods were proposed (Hu et al., 2021; Qian et al., 2021; Zhang et al., 2019), focusing on manually or automatically collecting relevant evidence from reliable data sources such as knowledge bases and fact-checking websites. However, due to the dynamic and high-volume nature of news events, these sources often struggle to update and verify information promptly, limiting the effectiveness of these methods. To address the challenges posed by a surge in new events, feature-based approaches have been introduced. These methods leverage deep learning to automatically extract event-agnostic features from news content and user comments, facilitating the identification of fake news. Feature-based approaches can be categorized into multimodal and single-modal methods. Multimodal methods (Jing et al., 2023; Luvembe et al., 2024; Ma et al., 2024; Song, Ning, et al., 2021; Ying et al., 2021) integrate features from multiple modalities, such as textual and visual data, to improve detection accuracy. While leveraging multiple modalities can enhance performance, news content does not always include visual elements and may appear solely in text form. In such cases, multimodal methods are not applicable, necessitating the development of robust single-modal text-based detection methods. Consequently, *a critical challenge in fake news detection research lies in designing effective strategies for extracting and representing discriminative features from the text of news content and user comments.*

1.1. Existing methods

To address the above challenge, current research in text-based fake news detection typically focuses on analyzing the content of news articles and their associated comments to assess their veracity (Ali et al., 2022; Kondamudi et al., 2023). A range of studies has advanced the extraction of feature representations from news content, leveraging elements such as semantic (Ahmad et al., 2025; Askarizade, 2025; Han et al., 2024; Hu et al., 2024; Pu et al., 2025; Qin & Zhang, 2024; Wang et al., 2025; Yan et al., 2024), structure (Karimi & Tang, 2019), and writing style (Park et al., 2025; Przybyla, 2020; Wu et al., 2024). Other studies have explored the use of comment features to improve detection performance, including comment semantics (Bazmi et al., 2023; Yang et al., 2023), user interaction patterns (Khoo et al., 2020; Mehta et al., 2022; Yan et al., 2024), propagation dynamics (Bian et al., 2020; Pröllochs et al., 2021), time evolution (Ruchansky et al., 2017; Song, Shu, & Wu, 2021), and user stance (Geng et al., 2024; Küçük & Can, 2020; Nguyen et al., 2020). In addition to these features, emotional signals have emerged as a powerful tool for distinguishing between real and fake news on social media (Pröllochs et al., 2021). A growing body of research has underscored the importance of emotion in detecting fake news from text (Ajao et al., 2019; Choudhry et al., 2024; Geng et al., 2024; Ghanem et al., 2021; Giachanou et al., 2019; Luvembe et al., 2023; Wu & Rao, 2020; Zhang et al., 2021, 2024). These studies suggest that emotional cues in news content and their associated comments can be indicative of its veracity. However, there is an aspect of fake news detection that is often left out in existing studies: the relationship between the emotions embedded in the content and the semantics of the comments, particularly those involving questioning versus non-questioning language. Therefore, we further investigate whether a discernible association exists between content emotion and comment semantics that could serve as a basis for distinguishing fake news from real news.

1.2. Research objective

We observe that fake news tends to induce distinct patterns of comment semantics compared to real news, depending on the emotion of the content. Specifically, we find that in comparison to real news, fake news content infused with emotions other than sadness tends to evoke questioning semantics while suppressing non-questioning semantics in comments. Conversely, fake news content infused with emotions of sadness has the opposite effect, which is more likely to suppress questioning semantics and instead elicit non-questioning semantics in comments. As illustrated in Fig. 1, for instance, fake news content injected with sad emotions causes 6.8% less questioning semantics in comments than real news (11.84% vs. 18.64%), which also means that it causes 6.8% more non-questioning semantics than real news. *These findings suggest that the interplay between content emotion and comment semantics is a critical factor in distinguishing real from fake news. However, existing methods tend to overlook this discriminative signal, thereby limiting their effectiveness. To address this gap, the objective of our study is to explore the latent associations between content emotion and comment semantics, particularly from the perspectives of questioning and non-questioning comments.*

1.3. Solution and contribution

In this paper, we propose an Emotion-Semantic Interaction Network (ESIN), which incorporates the interaction of content emotion with original comment semantics, as well as the distribution of comment semantic categories (i.e., questioning and non-questioning comments).

Specifically, we first propose two mechanisms in the ESIN, namely Codebook Initialization and Semantic Quantification, to capture the distribution of semantic categories in comments. The *Codebook Initialization* mechanism initializes the code (prototype) corresponding to each category of comment semantics and stores it in a codebook (Van Den Oord et al., 2017) through question filtering and non-question clustering. By extracting representative prototypes for each category, Codebook Initialization reduces the complexity of learning semantic information from a diverse set of comments and provides a foundation for the next step. The *Semantic Quantification* mechanism employs the initialized codebook to quantify the statistical distribution of comment semantics through code classification and counting. This distribution serves as a critical feature, enriching the interaction with content emotion and enhancing the model's ability to establish latent associations between content emotion and comment semantics.

Following this, the extracted semantic distribution and the original comment semantics undergo a cross-attention-based exchange with content emotions. This facilitates the acquisition of latent associations between content emotion and comment semantics, which are used as a representation of news veracity. Finally, this representation is input into a multi-layer perceptron for fake news detection.

The advantage and unique feature of the ESIN model compared with other models lies in that ESIN is question-aware, which innovatively distinguishes between questioning and non-questioning semantics in comments and facilitates a comprehensive interaction with the emotional cues embedded in news content. By leveraging its Codebook Initialization mechanism to capture representative prototypes for questioning and non-questioning comment categories, and Semantic Quantification to analyze their distribution, ESIN effectively models how content emotions influence user reactions. Through its cross-attention mechanism, the model captures the interplay between content emotion and comment semantics, providing a detailed and dynamic representation of their relationship. This approach sets ESIN apart from other methods, as it systematically integrates these underexplored dynamics to enhance the detection of fake news.

We conduct extended experiments on two widely-used fake news datasets, RumourEval (Gorrell et al., 2019) and WeiBo (Ma et al., 2016). Experimental results demonstrate that ESIN achieves competitive results, outperforming baseline models by a significant margin of 5.27% and 4.08% in weighted F1-score (F1.) and accuracy (Acc.) on the RumourEval dataset, and by 2.43% and 2.38% in F1. and Acc. on the WeiBo dataset. These promising results confirm the effectiveness of our model in distinguishing real from fake news based on the interaction between content emotion and comment semantics from the perspectives of questions and non-questions. Overall, this paper presents the following key contributions:

- We propose a question-aware Emotion-Semantic Interaction Network (ESIN) for fake news detection, which learns latent associations between content emotions and comment semantics, particularly focusing on the distinction between questioning vs. non-questioning comments.
- In particular, ESIN introduces two mechanisms, Codebook Initialization and Semantic Quantification, to capture the distribution of semantic categories in comments. Codebook Initialization extracts representative prototypes for different semantic categories (questioning vs. non-questioning comments), reducing the complexity of semantic learning from diverse comments. Semantic Quantification determines the distribution of these categories, providing key features for distinguishing fake from real news.
- The proposed ESIN model is comprehensively evaluated on two widely used fake news detection datasets, namely RumourEval (Gorrell et al., 2019) and WeiBo (Ma et al., 2016). Experimental results demonstrate that our ESIN achieves competitive performances on metrics of weighted average accuracy and F1-score on both datasets, thereby validating the effectiveness of our approach.

2. Related work

In addressing the challenge of text-based fake news detection, existing approaches can be broadly categorized into two types: those that disregard emotional factors (emotion-unrelated) and those that incorporate emotional analysis (emotion-related).

Emotion-unrelated methods mainly utilize semantics and other information except emotional signals to interact to determine the veracity of the news (Ahmad et al., 2025; Aljawarneh & Swedat, 2024; Askarizade, 2025; Bazmi et al., 2023; Bian et al., 2020; Chen et al., 2018; Guo et al., 2018, 2024; Han et al., 2024; Hassan et al., 2017; Hu et al., 2024; Jiang et al., 2022; Karimi & Tang, 2019; Khoo et al., 2020; Kim et al., 2019; Küçük & Can, 2020; Mehta et al., 2022; Nan et al., 2021; Nguyen et al., 2020; Park et al., 2025; Pérez-Rosas et al., 2018; Pröllochs et al., 2021; Przybyla, 2020; Pu et al., 2025; Qin & Zhang, 2024; Qu et al., 2020, 2020; Ruchansky et al., 2017; Sheng, et al., 2022; Shu et al., 2019; Song, Shu, & Wu, 2021; Wang et al., 2018, 2025; Wu et al., 2024; Yan et al., 2024; Yang et al., 2023).

A range of methods identify fake news by extracting features from news content. Specifically, some studies focus on extracting semantic (Ahmad et al., 2025; Aljawarneh & Swedat, 2024; Askarizade, 2025; Chen et al., 2018; Guo et al., 2018; Han et al., 2024; Hassan et al., 2017; Hu et al., 2024; Jiang et al., 2022; Kim et al., 2019; Nan et al., 2021; Pérez-Rosas et al., 2018; Pu et al., 2025; Qin & Zhang, 2024; Qu et al., 2020; Ruchansky et al., 2017; Sheng, et al., 2022; Shu et al., 2019; Wang et al., 2018, 2025; Yan et al., 2024) and structural features (Karimi & Tang, 2019) from news content to identify fake news. However, in addition to using semantic and logical disguises, fake news creators often also employ writing style disguises. While methods based on semantic structural features are suitable for style-neutral fake news disguised with semantic and logical features, they struggle to detect fake news disguised with style. To address this limitation, some studies proposed a series of fake news detection methods based on news writing features (Park et al., 2025; Przybyla, 2020; Wu et al., 2024). However, news content-based methods, including semantic structure and writing style, rely solely on the news text itself and fail to leverage news authenticity signals in comments, limiting their performance. To address this, a series of methods have emerged in recent years that analyze comment data to extract comment features related to news authenticity to enhance fake news detection. Specifically, relevant studies extract semantic information (Bazmi et al., 2023; Yang et al., 2023), structural features of news spread in comments (Bian et al., 2020; Pröllochs et al., 2021), and user interaction patterns (Khoo et al., 2020; Mehta et al., 2022; Yan et al., 2024) to assist in fake news detection. However, these methods focus on spatial features and ignore temporal evolution. Studies (Ruchansky et al., 2017; Song, Shu, & Wu, 2021) assist in fake news detection by analyzing and extracting temporal evolution features in comments. However, the above comment-based methods find it difficult to consider the impact of users' own biases (such as political inclinations). To this end, some methods (Geng et al., 2024; Küçük & Can, 2020; Nguyen et al., 2020) have introduced user stance and bias features to improve the performance of fake news detection. In short, features in comments have been widely used to assist in the judgment of news veracity, to improve the performance limitations of methods based solely on news content.

However, the above methods have largely overlooked the critical role of emotional factors. Existing literature analysis (Pröllochs et al., 2021) shows that emotional signals have a role in distinguishing true from false news. Leveraging such signals can improve the accuracy of news veracity judgments, yet most prior approaches neglect this aspect, limiting their effectiveness. To overcome this gap, researchers have introduced a range of emotion-related methods.

Emotion-related methods mainly extract and analyze emotional signals in news content and comments to enhance the performance of fake news detection (Ajao et al., 2019; Choudhry et al., 2024; Geng et al., 2024; Ghanem et al., 2021; Giachanou et al., 2019; Luvembe et al., 2023; Wu & Rao, 2020; Zhang et al., 2021, 2024). A series of studies (Ajao et al., 2019; Choudhry et al., 2024; Ghanem et al., 2021; Giachanou et al., 2019; Zhang et al., 2024) focuses on using emotional signals in news content to enhance the identification of fake news. Specifically, one approach examines the relationship between fake news and the sentiments conveyed in the news content (Ajao et al., 2019). Another method utilizes emotional signals from the text of claims, applying an LSTM neural network to differentiate between credible and non-credible claims (Giachanou et al., 2019). A model designed to track the flow of emotional information within fake news has been proposed to enhance detection accuracy (Ghanem et al., 2021). Graph attention networks (GATs) are also employed to capture emotional interactions across various types of articles, improving the detection process (Zhang et al., 2024). The connection between a text's emotional tone and its authenticity is explored in a multitasking framework that predicts both emotional content and the legitimacy of news articles for fake news detection (Choudhry et al., 2024). Studies (Geng et al., 2024; Wu & Rao, 2020; Zhang et al., 2021) further consider user emotions in comments, further enhancing the identification of fake news. Specifically, emotional associations and contradictions between the content and related comments are leveraged to identify fake news (Wu & Rao, 2020). Features including content emotion and comment emotion are integrated into the method, namely CEUS (Geng et al., 2024). In addition, differences in emotions expressed in news content and user comments are shown to help distinguish fake news from real news, with the introduction of a Dual Emotion Feature that strengthens detection performance when integrated into existing models (Zhang et al., 2021).

Although existing methods have advanced the field, they often treat semantic and emotional features separately, overlooking how a news content's emotion interacts with comment semantics, particularly questioning versus non-questioning language. Our investigation reveals that fake news elicits comment semantics that differ markedly from those of real news, with patterns shaped by the content's emotion. This interplay between content emotion and comment semantics emerges as a key signal for distinguishing real from fake news, yet it remains underutilized in current approaches, limiting their performance. To bridge this gap, our study aims to uncover the latent associations between content emotion and comment semantics, focusing specifically on questioning versus non-questioning comments.

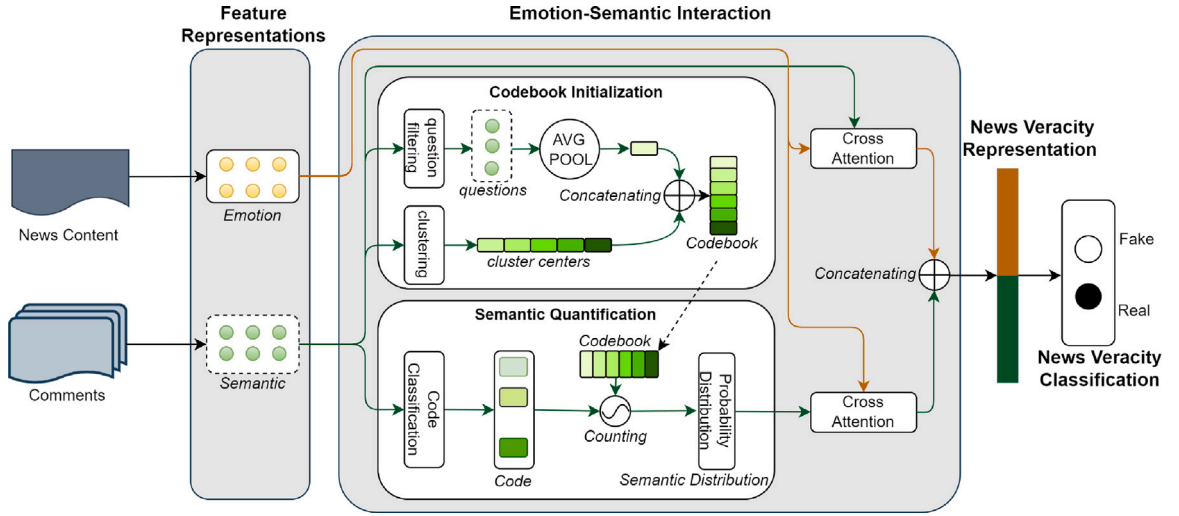


Fig. 2. The overall structure of our “question”-aware ESIN. First, the comment semantics associated with the news are fed into the mechanisms of Codebook Initialization and Semantic Quantification, which together generate the distribution of comment semantic categories. Next, the content emotion interacts with both the original comment semantics and the extracted distribution of comment semantic categories using cross-attention. The results of this interaction are concatenated to learn the latent relationship between content emotion and comment semantics, forming a news veracity representation. Finally, this news veracity representation is passed into a news veracity classifier, which infers the veracity (fake or real) of the news.

3. Methodology

3.1. Problem definition

We have a dataset $D = \{(news_i, y_i)\}_{i=1}^I$, where $y_i \in \{0, 1\}^2$ is the label of i th news representing the veracity of the news. I is the total number of news. We define $news_i = (x_{i,0}, x_{i,1}, \dots, x_{i,J})$ as the i th news, which is composed of the content $x_{i,0}$ and the comments $x_{i,j}$ of news. $j \in \{1, \dots, J\}$. J is the number of comments on each piece of news. We define each content $x_{i,0} = \{w_{i,0,0}, w_{i,0,1}, \dots, w_{i,0,L}\}$, and each comment $x_{i,j} = \{w_{i,j,0}, w_{i,j,1}, \dots, w_{i,j,L}\}$, which indicates that each content or comment on the i th news is composed of a sequence of L words. Our goal is to learn a function to infer the label y_i from news $news_i$.

3.2. Preprocess

In this work, we adopt the approach of Zhang et al. (2021), which provides a straightforward and reproducible framework with publicly available source code, to extract both emotion and semantic representations.

Emotion Representation: Following the prior work, we derive the emotion representation $E_{i,0}$ for the content of the i th news article, where $E_{i,0} \in \mathbb{R}^D$.

Semantic Representation: Similarly, we employ GloVe (Pennington et al., 2014) for English and Chinese Word Vectors (Li et al., 2018) for Chinese to obtain semantic representations from both news content and comments. This process yields a set of comment semantic representations $\{S_{i,j} \mid j \in \{1, \dots, J\}\}$ for the i th news article, where each $S_{i,j} \in \mathbb{R}^{L \times D}$.

3.3. Our ESIN

Our ESIN approach is illustrated in Fig. 2. In this approach, the emotional feature of the content interacts with the semantics of comments, which include both their original semantic features and the distribution of semantic categories. This interaction facilitates the discovery of latent associations between content emotion and comment semantics, forming a comprehensive representation of news veracity. This representation is then fed into a classifier to determine the veracity of the news.

Prior to the interaction between emotional and semantic information, we apply maximum pooling to the word representation vectors to derive the semantic representation vector at the comment level. As a result, we obtain the semantic representation vector $\bar{S}_{i,j}$ for each news comment, as denoted in Eq. (1).

$$\bar{S}_{i,j} = \text{Pool}(S_{i,j}) \quad (1)$$

where $S_{i,j} \in \mathbb{R}^{L \times D}$, and $\bar{S}_{i,j} \in \mathbb{R}^D$. L is the number of words in a comment.

Codebook Initialization: The goal of Codebook Initialization is to compute representative prototypes for the semantic categories of question/non-question comments. These prototypes are then used to quantify the original semantics of comments associated

with a news piece based on the codebook (Van Den Oord et al., 2017). Specifically, we adopt the codebook because each semantic category encompasses numerous comments with varied expressions, which introduces significant variance. To address this, representative prototypes are extracted for each category to simplify the learning process of semantic category information and facilitate the quantification of semantic distributions. Inspired by the idea behind codebooks, originally designed to extract discrete representations of images (Van Den Oord et al., 2017), we strategically configure the codebook to store these representative representations for each comment category effectively.

To construct the codebook, we begin by addressing the semantic representation of question comments. Specifically, we calculate the average semantic vector of all question comments containing question marks (“?”) in the training set. This is represented as $\{\bar{S}_{i,j} | i \in \{1, \dots, I\}, j \in \text{indexes of question comments}\}$, resulting in the vector $code_q \in \mathbb{R}^{1 \times D}$, which serves as the prototype for the question category. This vector is then added to the codebook.

Beyond question-based semantics, comments may express a range of other non-question semantic types that need to be adequately captured (such as exclamatory, declarative, or other non-question expressions). Rather than treating all non-question comments as a single, homogenous group, we aim to represent the diverse latent categories within them. To achieve this, we apply clustering to the semantic vectors of non-question comments, denoted as $\{\bar{S}_{i,j} | i \in \{1, \dots, I\}, j \in \text{indexes of non-question comments}\}$. This process identifies latent categories, generating prototypes $code_{nq} \in \mathbb{R}^{(C-1) \times D}$, where each vector corresponds to a distinct latent semantic category of non-question comments. Here, $C-1$ represents the predefined number of latent categories. The decision to use clustering stems from the assumption that comments belonging to the same latent category are more similar in semantic space. We employ K-means clustering (Hartigan & Wong, 1979) to determine the cluster centers, which then serve as the prototypes $code_{nq}$ for the latent semantic categories. These cluster centers are subsequently added to the codebook.

Finally, the complete codebook is formed by concatenating $code_q$ with $code_{nq}$, resulting in $Codebook = \text{Concat}(code_q, code_{nq})$, where $Codebook \in \mathbb{R}^{C \times D}$ and C is the total number of codes in the codebook. We define $c \in \{1, 2, \dots, C\}$, such that when $c > 1$, $code^c$ belongs to $code_{nq}$.

Semantic Quantification: The objective of Semantic Quantification is to leverage the initialized codebook to measure the statistical distribution of comment semantics through code-based classification and frequency counting.

In this process, we begin by classifying each comment into a specific code from the initialized codebook. This step discretizes the semantic representation of comments, associating each comment with its corresponding prototype code. Subsequently, we count the occurrences of these codes within a given piece of news, calculating the frequency of each code. This frequency represents the distribution of semantic categories in the comments associated with the news.

Through this process, termed Semantic Quantification, we effectively translate the semantics of comments into a quantifiable distribution based on their alignment with the codebook prototypes. Formally, these processes are defined in Eqs. (2) and (3).

$$CodeID_{i,j} = \begin{cases} \arg \min_c \{\|\bar{S}_{i,j} - code^c\|_2 | c \neq 1\} & \text{for non-questions} \\ 1 & \text{for questions} \end{cases} \quad (2)$$

$$Dist_i = \text{Count}(\{CodeID_{i,j} | j > 0\}) \quad (3)$$

where $Dist_i \in \mathbb{R}^C$.

Specifically, for non-questioning comments, we calculate the similarity between their semantic vectors $\bar{S}_{i,j}$ and the codes representing non-questioning categories using the Euclidean distance. Each $\bar{S}_{i,j}$ is then assigned to the code with the smallest Euclidean distance, determining its closest match.

For questioning comments, we directly assign the predefined code $code_q$ to all corresponding comments without further calculation.

As a result, we obtain the code ID $CodeID_{i,j} \in \{1, \dots, C\}$ for each comment, as described in Eq. (2). Finally, we count the occurrences of each code ID in the i th piece of news, producing the distribution of comment semantic categories, denoted as $Dist_i$, as defined in Eq. (3).

Information Interaction: The goal of Information Interaction is to jointly infer the emotion by integrating the extracted comment semantic category information with the content emotion. Below, we outline the process of interaction between content emotion and comment semantics.

First, the distribution of comment semantic categories, obtained through the previously described process, is used to attend to the content emotion, dynamically learning a latent distribution R_i^1 . Next, the content emotion attends to the original comment semantics, enriching the contextual information within the content emotion and generating R_i^2 . These two results, R_i^1 and R_i^2 , are then concatenated to capture the latent relationships between content emotion and comment semantics, forming the news veracity representation R_i , as defined in Eq. (4).

Note that the “attend” operation in the above steps refers to the CrossAttention function, whose implementation is detailed in Eq. (5), Eq. (6), and Eq. (7).

$$\begin{aligned} R_i^1 &= \text{CrossAttention}(Dist_i, E_{i,0}W_e, E_{i,0}W_e) \\ R_i^2 &= \text{CrossAttention}(E_{i,0}, S_i, S_i) \\ R_i &= \text{Concat}(R_i^1, R_i^2) \end{aligned} \quad (4)$$

where $W_e \in \mathbb{R}^{D \times C}$, $R_i^1 \in \mathbb{R}^C$, $R_i^2 \in \mathbb{R}^D$, $R_i \in \mathbb{R}^{C+D}$. In particular, W_e serves as a dimensional alignment transformation matrix, designed to facilitate attention interaction by aligning dimensions.

The standard CrossAttention function is defined in Eq. (5), Eq. (6), and Eq. (7).

For the function $\text{CrossAttention}(q, k, v)$, where $q \in \mathbb{R}^{L_q \times D}$, $k \in \mathbb{R}^{L_k \times D}$, $v \in \mathbb{R}^{L_v \times D}$, we compute a group of queries, keys, and values.

$$\begin{aligned} Q &= W_q(q)^T \\ K &= W_k(k)^T \\ V &= W_v(v)^T \end{aligned} \quad (5)$$

where $W_q, W_k \in \mathbb{R}^{D_k \times D}$, $W_v \in \mathbb{R}^{D_v \times D}$, $Q \in \mathbb{R}^{D_k \times L_q}$, $K \in \mathbb{R}^{D_k \times L_k}$, $V \in \mathbb{R}^{D_v \times L_v}$. Among them, W_q , W_k , and W_v are linear transformation matrices corresponding to queries, keys, and values, respectively, enabling adaptable representations for attention mechanisms. After that, one-head cross-attention is computed.

$$(H)^1 = \text{Softmax}\left(\frac{(Q)^T K}{\sqrt{D}}\right)(V)^T \quad (6)$$

where $(H)^1 \in \mathbb{R}^{L_q \times D_v}$ and $(\cdot)^1$ indicates the first head. Here, multi-head attention is applied for diverse representation learning. Formally, for h heads,

$$\hat{q} = (W_{head}(\text{Concat}((H)^1, \dots, (H)^h))^T)^T \quad (7)$$

where $W_{head} \in \mathbb{R}^{D \times (h \times D_v)}$, $\hat{q} \in \mathbb{R}^{L_q \times D}$. As a result, we obtain a latent representation matrix of q , that is, $\hat{q}$.

Finally, in the process of information interaction, we obtain the news veracity representation R_i .

News Veracity Classification: The news veracity representation R_i derived from the previous processes serves as the foundation for inferring the veracity of the news (fake or real). To achieve this, we employ a multi-layer perceptron (MLP). The representation R_i is fed into the MLP, which outputs a probability distribution over the possible veracity labels of the news. This process is formally defined in Eq. (8).

$$P_i = \text{Softmax}(\text{MLP}(R_i)) \quad (8)$$

where P_i is the probability distribution of news veracity inferred by the news veracity representation R_i . Here, MLP is the multi-layer perceptron, which is defined in Eq. (9).

$$\text{MLP}(x) = (\text{ReLU}(xW_1 + b_1))W_2 + b_2 \quad (9)$$

where $W_1 \in \mathbb{R}^{D \times D_h}$, $b_1 \in \mathbb{R}^{D_h}$, $W_2 \in \mathbb{R}^{D_h \times 2}$, $b_2 \in \mathbb{R}^2$. W_1 is the weight matrix of the first layer, b_1 is the bias vector of the first layer, adding offsets to each neuron in the hidden layer, W_2 is the weight matrix of the second layer, mapping the hidden layer from the hidden layer dimension to the output dimension, b_2 is the bias vector of the second layer, adding offsets to each neuron in the output layer.

Objective Function: Finally, we employ cross-entropy to compute the error between the predicted probability distribution and the ground truth of news veracity, which serves as the objective function. To optimize this function, we use a backpropagation-based optimization algorithm, such as Adam (Kingma & Ba, 2014), to update the model parameters. The objective function is formally defined in Eq. (10):

$$\mathcal{L} = -\frac{\sum_{i=1}^I \log P_i[y_i]}{I} \quad (10)$$

where I is the number of samples, P_i is the predicted probability distribution for the i th piece of news, and y_i is the ground-truth label indicating the veracity of the i th news.

4. Experiments

4.1. Experimental setup

4.1.1. Datasets

We conduct experiments using two widely utilized fake news datasets: RumourEval (Gorrell et al., 2019) in English and WeiBo (Ma et al., 2016) in Chinese. These datasets are well-suited for validating our method as they encompass both news content and news comments. The statistics of the two datasets used for our experiments are shown in Table 1.

RumourEval: The dataset RumourEval is created to determine the veracity of news pieces on Twitter and Reddit, which is a popular fake news detection dataset in English. We use the news that is labeled as fake or real in the dataset. We follow the official setting to split the dataset.

WeiBo: The WeiBo dataset is a popular fake news detection dataset in Chinese, where each news piece is labeled as fake or real. We de-duplicate the news in this dataset for data cleaning and keep the de-duplicated dataset for our experiments. Finally, we split the cleaned dataset into the training set, validation set, and test set with a ratio of 3:1:1.

Table 1

The statistics of datasets. # cnt: the number of pieces of news content. # cmt: the number of news comments.

		RumourEval		Weibo	
		# cnt	# cmt	# cnt	# cmt
Train	# Fake	79	1135	801	649 673
	# Real	144	1905	1410	482 226
	# Total	223	3040	2211	1 131 899
Valid	# Fake	19	824	268	222 149
	# Real	10	404	470	146 948
	# Total	29	1228	738	369 097
Test	# Fake	40	689	286	193 740
	# Real	31	805	471	179 942
	# Total	71	1494	757	373 682
Total	# Fake	138	2648	1355	1 065 562
	# Real	185	3114	2351	809 116
	# Total	323	5762	3706	1 874 678

4.1.2. Baselines

To evaluate the effectiveness of our ESIN, we selected sixteen competitive fake news detectors as baselines. These detectors have publicly available original code and align with our problem definition, ensuring a fair comparison.

BiGRU & BiGRU-EMO: The RNN-based models have been shown to be effective for the fake news detection task (Chen et al., 2018; Giachanou et al., 2019; Guo et al., 2018; Ma et al., 2016; Ruchansky et al., 2017). Therefore, we choose a representative RNN model BiGRU as one of the baselines and take the semantic representations of news content and news comments as the input of the model. Furthermore, to incorporate emotional information into the baselines for a fairer comparison with our ESIN, which leverages emotion, we integrate the Dual Emotion Feature (DEF) (Zhang et al., 2021). DEF is open-sourced, straightforward to implement, and has been shown to effectively enhance fake news detection performance. Specifically, we embed DEF into the BiGRU model to create a baseline called BiGRU-EMO, ensuring that emotional information is leveraged. This setup allows us to test the effectiveness of our proposed approach against a competitive baseline while maintaining consistency in the evaluation process.

EANN & EANN-EMO: Furthermore, we compare our model with EANN (Wang et al., 2018), a well-established and competitive baseline in the fake news detection domain, to evaluate the effectiveness of our proposed approach. In the same vein, we incorporate the DEF into EANN, resulting in a modified baseline model called EANN-EMO, which is then trained to determine the veracity of the news.

InfOp & InfOp-EMO: We conduct a comparison of our model with InfOp (Mehta et al., 2022). In this comparison, we specifically utilize the user–user inference operators in InfOp to enhance the semantic representations of comments through a graph convolutional network, ensuring that both the news content and comments are considered for a balanced analysis. Additionally, we introduce DEF into InfOp to create a variant named InfOp-EMO, which is then trained to assess the veracity of the news.

MDFEND & MDFEND-EMO: We compare our model with MDFEND (Nan et al., 2021), a proven fake news detection method that employs a domain gate to combine multiple representations derived from a mixture of experts with attention mechanisms. This model effectively aggregates diverse features to enhance detection accuracy. To ensure a fair evaluation, both news content and comments are included in the comparison. In a similar manner, we incorporate DEF into MDFEND to create a modified baseline, MDFEND-EMO, which is then trained to predict the veracity of the news.

DualCAN & DualCAN-EMO: We evaluate our model by comparing it with DualCAN (Yang et al., 2023), a comprehensive approach that incorporates news content, comments, and external knowledge to improve fake news detection. This method has shown promise by leveraging multiple sources of information for more accurate predictions. Similarly, we integrate DEF into DualCAN, creating a modified baseline model, DualCAN-EMO, and train it to detect fake news.

ARG & ARG-EMO: We compare our model against ARG (Hu et al., 2024), which is a state-of-the-art (SOTA) fake news detection model. Both news content and news comments are utilized for a fair comparison. Similarly, we plug DEF into ARG as the baseline ARG-EMO and then train the model to detect fake news.

SheepDog & SheepDog-EMO: We compare our model against SheepDog (Wu et al., 2024), which is also an effective SOTA fake news detection model. Both news content and news comments are considered. Similarly, we integrate DEF into SheepDog, creating a baseline named SheepDog-EMO, and subsequently train this model to identify fake news.

CEUS: We compare our model with CEUS (Geng et al., 2024), a state-of-the-art (SOTA) model for fake news detection that incorporates features such as news content emotion, comment emotion, comment semantics, and stance. Given that CEUS already leverages news content emotion, we have refrained from incorporating additional emotion signals into our model.

L-E-MTL: We compare our model with the best of the SOTA models implemented (Choudhry et al., 2024), which employs the BERT-base model as its backbone to predict both the legitimacy and emotion of news, referred to here as L-E-MTL by us. As an emotion-aware method, L-E-MTL leverages emotional information through bi-task supervision, facilitating implicit interactions between emotional cues and news content during the training phase. Given that this approach inherently captures the relationship between emotion and content, we do not further incorporate DEF into this approach. Similarly, both news content and comments are utilized.

Table 2

Performance comparison of our ESIN against sixteen baselines. In each group, the one that worked best is marked in bold. F1.=weighted average F1-score. Acc.=weighted average accuracy.

Methods	RumourEval		WeiBo	
	F1.	Acc.	F1.	Acc.
BiGRU	48.27	49.44	76.48	77.74
EANN	51.53	51.97	72.30	72.83
InfOp	47.12	47.32	75.84	77.52
MDFEND	49.22	52.10	53.56	60.28
DualCAN	48.73	52.11	77.35	77.41
ARG	44.96	49.31	88.56	88.82
SheepDog	51.03	53.50	88.46	89.30
BiGRU-EMO	52.67	53.32	79.23	79.39
EANN-EMO	50.86	51.13	76.77	77.89
InfOp-EMO	43.46	43.52	75.29	77.07
MDFEND-EMO	51.42	54.65	53.46	54.69
DualCAN-EMO	49.46	54.09	79.02	79.26
ARG-EMO	51.45	52.37	89.28	89.70
SheepDog-EMO	52.15	53.52	88.44	89.04
CEUS	49.20	52.20	90.31	90.41
L-E-MTL	50.33	52.11	85.72	86.92
ESIN(Ours)	57.94	58.73	92.74	92.79

4.1.3. Settings

The number of words L in each content or comment is set to 100. The dimension of emotional representation and semantic representation is set to $D = 200$. The number of codes C in the codebook is set to 4 on the RumourEval and WeiBo datasets, which are tried in the range of 2 to 6 (see Section 4.5). For Cross-Attention in Eq. (5), Eq. (6), and Eq. (7), we set D_q , D_k and D_v to be 64, the number of heads h is set to 8. For the fully connected layer in Eq. (9) for news veracity classification, we set D_h to 32. We utilize the Adam (Kingma & Ba, 2014) optimizer to train our model. For the RumourEval dataset, we set the learning rate to 1e-3 and the epoch to 50. For the WeiBo dataset, we set the learning rate to 1e-5 and the epoch to 300. We set the batch size to 16 on both datasets. The Wilcoxon rank test was used in the experiments to verify statistical significance (p value < 0.05). Our experiments are conducted on a GPU device (NVIDIA A100 80G).

4.2. Model comparison

To validate the effectiveness of ESIN, we compare its performance against sixteen baselines on the RumourEval and WeiBo datasets. Evaluation metrics include weighted average F1-score (F1.) and accuracy (Acc.), both of which are well-suited for binary classification tasks like fake news detection. The experimental results are shown in Table 2, with the best performance for each metric highlighted in bold. For each column, the entry in bold represents the result that performs better, with a p -value less than 0.05 indicating statistical significance. We conducted 10 rounds of experiments and report the average results for the F1 and Accuracy metrics to evaluate ESIN. As shown in Table 2, ESIN consistently outperforms the sixteen baselines, achieving improvements of 5.27% (57.94% vs. 52.67%) and 4.08% (58.73% vs. 54.65%) in F1. and Acc. on the RumourEval dataset, and 2.43% (92.74% vs. 90.31%) and 2.38% (92.79% vs. 90.41%) in F1. and Acc. on the WeiBo dataset.

These results highlight the effectiveness of our approach in leveraging the interaction between content emotion and comment semantics from questioning and non-questioning perspectives. By specifically focusing on the “Questions” and “Non-Questions” in comment semantics, ESIN can capture these differences more effectively than baseline models. Baselines that do not explicitly consider the role of questioning/non-questioning semantics may miss these crucial cues. For instance, models that only focus on overall semantic features or emotional signals in a general way may not be able to distinguish between the questioning patterns in fake and real news. ESIN, however, through its Codebook Initialization and Semantic Quantification mechanisms, can accurately extract and analyze the frequency of questioning and non-questioning comments. This allows the model to better learn the latent associations between content emotions and comment semantics related to questioning and non-questioning, thereby improving the accuracy of fake news detection. In conclusion, the significant improvements achieved by ESIN, coupled with the insights derived from our experiments, validate the effectiveness of our approach for fake news detection. These results emphasize the importance of exploring latent associations between content emotion and comment semantics, especially those related to questioning and non-questioning, for this task. The unique consideration of “Questions”/“Non-Questions” in our ESIN model provides a new perspective and a more accurate way to identify fake news in a social media environment.

4.3. Ablation study

The ablation study aims to evaluate the importance of the following components in ESIN: (1) the extracted distribution of the questioning category in comments, (2) non-questioning semantic clustering, (3) the overall distribution of semantic categories, and (4) interaction with content emotions. To this end, we individually remove the corresponding mechanisms (Question Filtering,

Table 3
Ablation study of four key components in our ESIN on two datasets.

Dataset	Ablation	F1.	Acc.
RumourEval	ESIN	57.94	58.73
	w/o Question Filtering	55.29	56.06
	w/o Semantic Clustering	56.04	56.65
	w/o Semantic Quantification	52.72	53.66
	w/o Content Emotion	42.19	42.68
WeiBo	ESIN	92.74	92.79
	w/o Question Filtering	91.57	91.65
	w/o Semantic Clustering	91.67	91.75
	w/o Semantic Quantification	91.36	91.44
	w/o Content Emotion	80.39	80.83

Semantic Clustering, Semantic Quantification, and Content Emotion) and assess their impact on F1. and Acc. using the RumourEval and WeiBo datasets. The results, presented in Table 3, demonstrate a significant decrease in ESIN's performance on F1. and Acc. metrics when any of these components is removed. The impact of each component, ranked from most to least critical, is as follows: Content Emotion, Semantic Quantification, Question Filtering, and Semantic Clustering.

Without Content Emotion, comment semantics and their distributions cannot interact with content emotion. This prevents the model from learning the latent associations between these elements, leading to a substantial drop in performance for news veracity inference. On the RumourEval dataset, the F1. score plummets to 42.19, and the Acc. score drops to 42.68. On the WeiBo dataset, the F1. score and Acc. score also show significant declines, reaching 80.39 and 80.83, respectively.

Semantic Quantification is a key process that calculates the distribution of all semantic categories, including questioning and non-questioning. It uses the initialized codebook to classify comments into different codes and then counts the frequency of each code in the news. This process provides a comprehensive statistical view of the comment semantics, enabling the model to better understand the overall semantic distribution in relation to the news content. When Semantic Quantification is removed, the model is unable to extract these key statistical features related to comment semantic distributions. This omission makes it difficult for the model to capture the distribution information across diverse comment expressions, resulting in a marked decrease in performance. On the RumourEval dataset, the F1 score drops to 52.72, while the accuracy score decreases to 53.66. Similarly, on the WeiBo dataset, both the F1 score and accuracy score experience notable declines, falling to 91.36 and 91.44, respectively.

When we remove the Question Filtering mechanism in the ablation study, the model loses its ability to specifically identify and analyze the questioning category. As a result, the quality and completeness of the distribution information related to questioning comments are severely reduced. Without this crucial information, the model struggles to capture the unique patterns of fake news that often trigger a high frequency of doubt-expressing questions. This leads to a significant decrease in performance, as shown in Table 3. On the RumourEval dataset, the F1. score drops from 57.94 to 55.29, and the Acc. score drops from 58.73 to 56.06. On the WeiBo dataset, the F1. score decreases from 92.74 to 91.57, and the Acc. score drops from 92.79 to 91.65. In summary, the removal of the frequency calculation of “questions”/“non-questions” has a negative impact on the performance of the model.

In addition, when Semantic Clustering is removed, the granularity of non-questioning semantics is diminished. This limits the diversity of distribution information available for interaction with content emotions, thereby reducing the model's ability to distinguish fake news effectively. On the RumourEval dataset, the F1 score decreases to 56.04, while the accuracy score drops to 56.65. Similarly, on the WeiBo dataset, both the F1 score and accuracy score see significant declines, falling to 91.67 and 91.75, respectively.

In conclusion, the ablation study clearly demonstrates the critical role of news emotion, the Question/Non-Question categorization, and distribution calculation in enabling ESIN to achieve robust performance. Their combined functionality ensures the effective modeling of interactions between content emotions and comment semantics, which is essential for accurate fake news detection. Each component, whether it is the identification of questioning comments, the clustering of non-questioning semantics, the quantification of semantic distributions, or news content emotion, contributes uniquely to the model's ability to distinguish between real and fake news.

4.4. Impact of the number of news and comments

In our experiments, as shown in Table 2, we observed that most models performed significantly better on the WeiBo dataset, which contains a larger volume of news and comments, compared to the RumourEval dataset, which has fewer samples. This disparity prompted us to investigate how the performance of our model changes when the size of the WeiBo dataset is reduced, particularly in terms of the number of news articles and comments available.

To understand the impact of dataset size on model performance, we first reduced the sample size of news articles from the WeiBo dataset by randomly selecting 70%, 40%, and 10% of the original dataset. The results, shown in Fig. 3(a), reveal a clear trend: performance is positively correlated with the sample size. Specifically, as the number of news articles decreases, both the F1 score and the accuracy of the model decrease. This behavior can be attributed to the limited number of news articles, which increases the model's estimation error regarding the true classification boundary. With fewer examples, the model struggles to generalize and correctly identify the underlying patterns in the data, leading to a drop in performance.

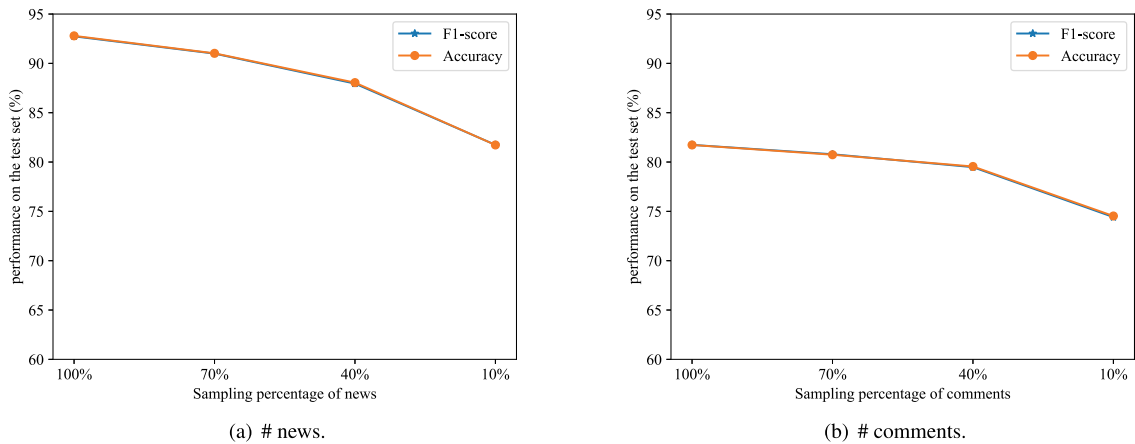


Fig. 3. Analysis of the number of news and comments.

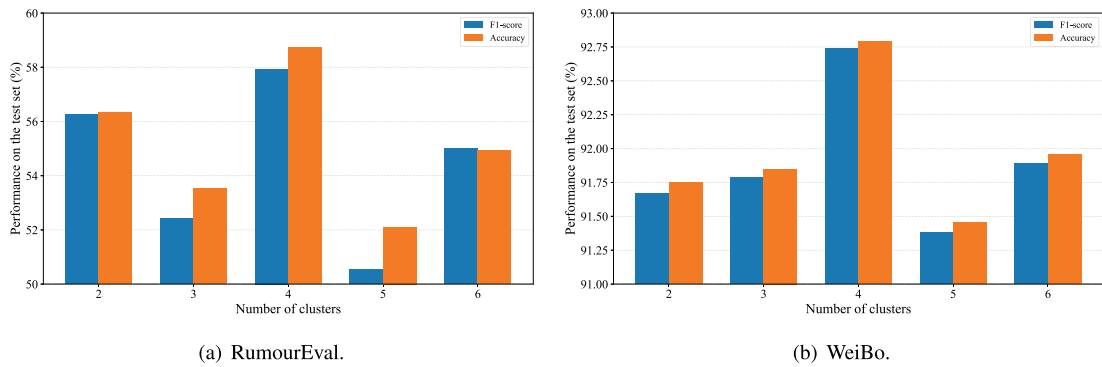


Fig. 4. Analysis of the key hyper-parameter: the number of clusters.

Next, we further reduced the number of comments under the condition that only 10% of the news articles remained. We sampled 70%, 40%, and 10% of the original comments and observed the results in Fig. 3(b). As expected, when the number of comments decreases, the model's performance deteriorates further. This drop in performance can be attributed to the increased estimation error in the model's understanding of the probability distribution of semantic categories (questioning vs. non-questioning comments). With fewer comments, the sample size for each semantic category becomes smaller. This leads to a situation where the model may not have enough data points to accurately estimate the probability distribution of questioning and non-questioning comments. As a result, the model's ability to accurately represent these semantic categories in the codebook is weakened, which negatively impacts the overall performance. In summary, when the number of comments decreases, the ability of the frequency of "questions" to represent the distribution of "questions" weakens, which negatively impacts the performance of the model.

An additional observation was made when the sample size of both news articles and comments from WeiBo was reduced to 10%, bringing it closer in size to the RumourEval dataset. In this case, the performance on WeiBo became more similar to that on RumourEval, but still slightly better. We hypothesize that this discrepancy is due to the differences in linguistic expression between Chinese (WeiBo) and English (RumourEval), which may introduce some performance fluctuations.

In summary, these experiments validate the monotonic relationship between the number of news articles and comments and model performance.

4.5. Impact of the number of clusters

The key hyperparameter, that is, the number of clusters C of Semantic Clustering, is analyzed by changing the number of clusters C in the range of 2 to 6, as shown in Fig. 4. This figure shows how the performance of the model ESIN changes when the number of clusters C varies. From this figure, we find that ESIN consistently achieves competitive performance against baselines with varying numbers of clusters, and achieves the best results on RumourEval and WeiBo when C is equal to 4.

We also visualize the word distribution in clusters of comment semantic categories, to further understand the information in each category after semantic clustering, as shown in Figs. 5 and 6. We can see the symbol "?" has a large frequency in Fig. 5(a). The number of comments in the c1 category is the least, and it is full of the word "its" as shown in Fig. 5(b). The number of comments in

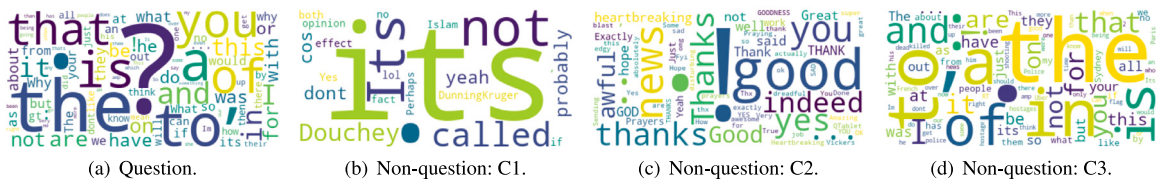


Fig. 5. Word clouds of comment semantic categories (including the questioning category and the latent categories C1, C2, and C3) in the RumourEval dataset.

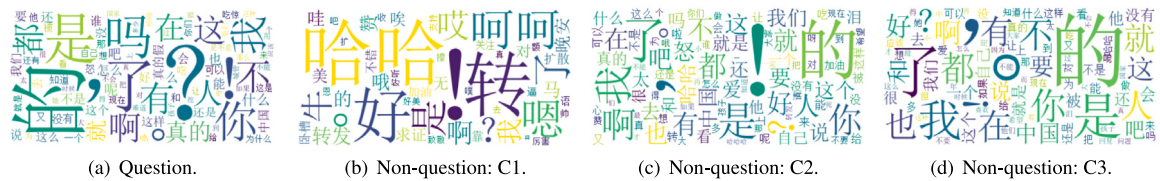


Fig. 6. Word clouds of comment semantic categories in the WeiBo dataset.

the c2 category is full of exclamation words, such as “good”, “thanks”, and “!”, etc., as shown in Fig. 5(c). The c3 category in Fig. 5(d) is basically declarative sentences. We can find that through semantic clustering, we divide the non-questioning category into a more fine-grained outlier category c1 with little information, an exclamation category c2, and a declarative category c3. This provides support for enhancing the statistical information of comment semantic categories. A similar phenomenon can also be observed in Fig. 6, dividing the non-questioning category into a more fine-grained outlier category c1 with little information separated from exclamatory sentences, exclamation category c2, and declarative category c3. The clustering results further corroborate the effectiveness of our approach in isolating different types of comment semantics. The clear distinction between these categories, particularly the fine-grained nature of the clusters, underscores the power of semantic clustering in capturing diverse comment behaviors and improving the accuracy of fake news detection. Overall, through this visualization, we can see how semantic clustering works.

4.6. Case study

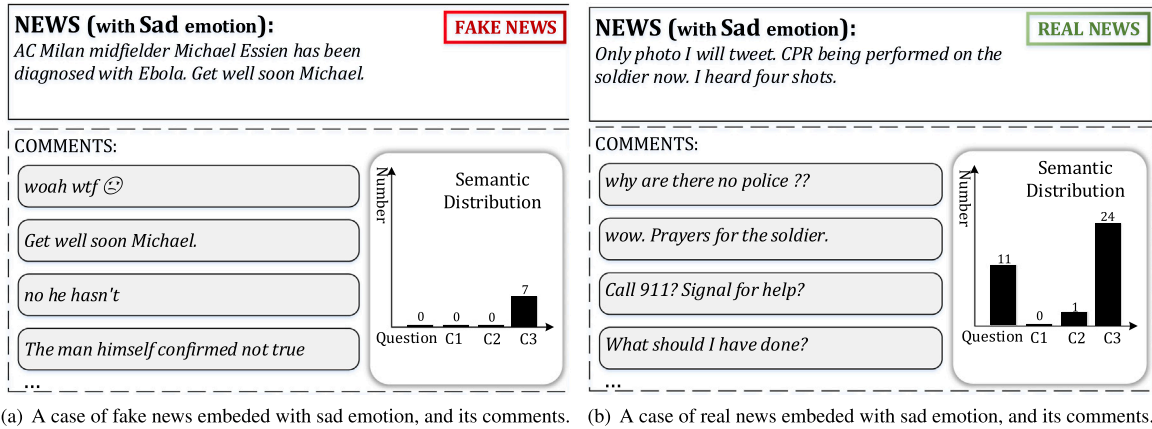
To further illustrate how our ESIN functions, we visualize two pairs of news cases that our model successfully detects, as shown in Figs. 7 and 8. The goal of this visualization is to offer a clearer understanding of how ESIN distinguishes between real and fake news by analyzing the distribution of comment semantics based on the emotional content of the news.

In the first pair of news cases, shown in Figs. 7(a) and 7(b), both pieces of news contain sad emotions. One of these is real news, while the other is fake news. Through ESIN’s semantic distribution analysis, we observe distinct differences in the comment semantics for these two cases. For the fake news case (Fig. 7(a)), which is embedded with sad emotions, the semantic distribution reveals that no comments fall into the questioning category, and only latent categories c1 and c2 (which represent specific non-questioning semantics) have 0 occurrences, with the majority of comments falling into category c3. This indicates that sad fake news does not trigger questioning semantics in the comments. In contrast, for the real news case with a similar emotional tone (Fig. 7(b)), we see a marked difference: 11 comments are categorized as questioning semantics, and there is a distribution across latent categories c1 (0), c2 (1), and c3 (24). In the second pair of news cases (Figs. 8(a) and 8(b)), both pieces of news carry joy as the emotional tone. Again, one news piece is fake, and the other is real. For the fake news case (Fig. 8(a)) with joy as the content emotion, we observe a higher frequency of questioning comments compared to the real news case (Fig. 8(b)).

These results confirm that the emotional tone of the news plays a significant role in shaping the distribution of comment semantics. ESIN’s ability to capture these latent associations between content emotion and comment semantics makes it particularly effective at distinguishing between real and fake news, as it leverages not only the content of the news itself but also the patterns of interaction in the comments.

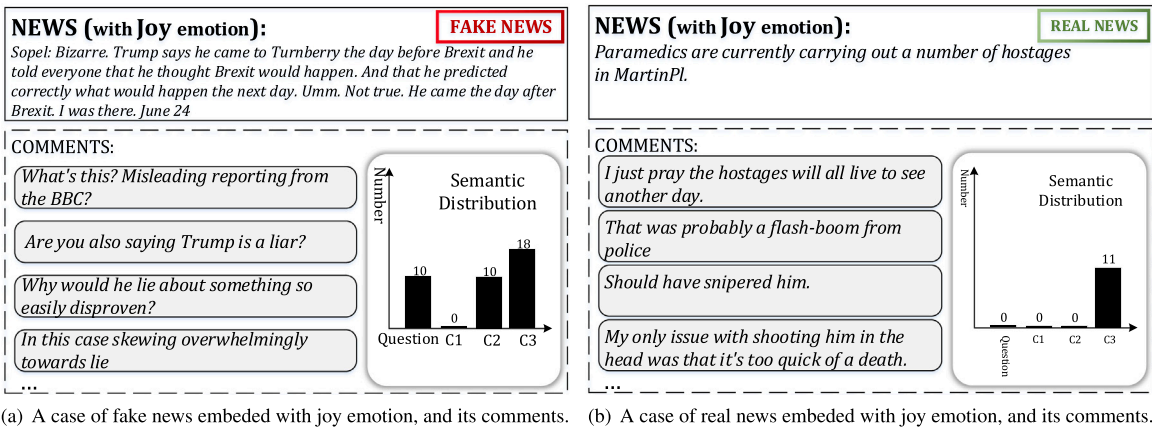
To provide a deeper analysis of the differences in the semantic distribution of comments on real and fake news, we visualized the distribution of all news samples across two dimensions: questioning and non-questioning semantics. The results are presented in Fig. 9. Observing Fig. 1, we noted that the questioning patterns of news with Sadness in their content differ significantly from those associated with other emotions. Consequently, we distinguished Sadness from non-Sadness emotions in our visualizations. Fig. 9(a) illustrates the semantic distribution of comments corresponding to news with non-Sadness emotions. Fig. 9(b) presents the semantic distribution for news characterized by Sadness. The horizontal axis represents the number of non-questioning comments, and the vertical axis represents the number of questioning comments. Red dots indicate fake news, while green dots denote real news. To better highlight the differences, we plotted two oblique lines representing the overall questioning tendencies for real and fake news comments. These lines were drawn using the average ratio of question volume to non-question volume as the slope.

The visualizations reveal statistically significant differences in the semantic category distribution of comments between fake news and real news (red vs. green dots). For non-Sadness news (Fig. 9(a)): The semantic distribution of comments for fake news tends to



(a) A case of fake news embedded with sad emotion, and its comments. (b) A case of real news embedded with sad emotion, and its comments.

Fig. 7. Visualization of a pair of news cases in the RumourEval dataset, one is fake news and the other is real news, both of which have Sad emotions in their contents.



(a) A case of fake news embedded with joy emotion, and its comments. (b) A case of real news embedded with joy emotion, and its comments.

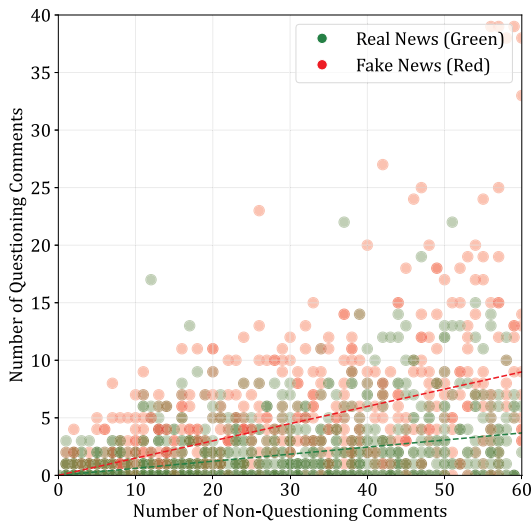
Fig. 8. Visualization of a pair of news cases in the RumourEval dataset, one is fake news and the other is real news, both of which have Joy emotions in their contents.

drift toward the top of the figure. For Sadness news (Fig. 9(b)), the trend is reversed, probably due to empathy and sympathy in readers. These observations demonstrate that the differences in semantic distributions between true and false news are not limited to isolated cases but are a more general phenomenon, reinforcing the idea that this distinction is both significant and prevalent.

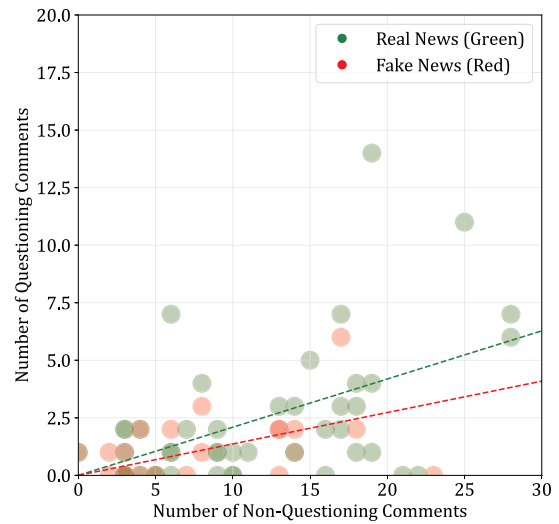
To further analyze how these semantic distributions affect the ESIN model, we evaluated its performance across different questioning frequency bands. Based on the questioning frequency differences observed in Sadness news in Fig. 1 (18.64% for fake news vs. 11.84% for real news), we set a 15% frequency threshold as it is roughly the midpoint between the two values. We then compared the model's predictions on news samples with questioning frequencies below and above this threshold, separately examining Sadness and non-Sadness categories. The experimental results are illustrated in Fig. 10, revealing the following key observations:

(1) Regardless of the emotion bin (Not Sad or Sad), the model exhibits statistically significant differences in prediction tendencies across questioning frequency intervals. For example, for the Not Sad bin, when the questioning frequency is low, the model tends to predict the news as real. Conversely, when the questioning frequency is high, the model is more inclined to predict the news as fake. This indicates that the ESIN model is influenced by the distribution of questioning comments. It effectively learns patterns for distinguishing news veracity based on questioning frequencies and adjusts its predictions accordingly across different frequency bands.

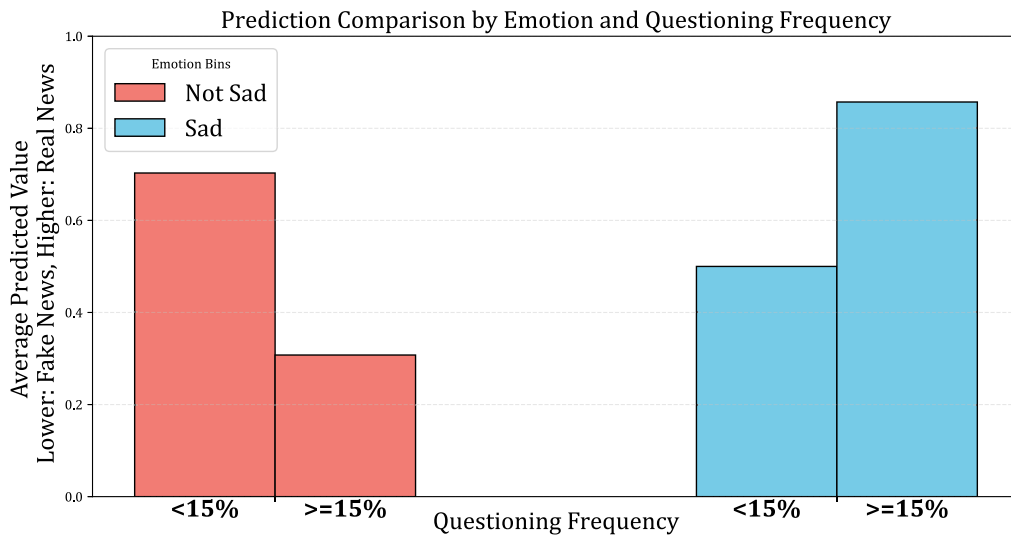
(2) The model's prediction behavior varies notably between the two emotion bins. In the Not Sad bin, the model is more likely to classify news with lower questioning frequencies as real. In the Sad bin, the model demonstrates the opposite tendency. These patterns align with the trends observed in Fig. 1, where the relationship between questioning frequency and news veracity differs based on the emotions of the news content. These findings highlight the model's capacity to effectively capture and leverage patterns within the data. Specifically, the model integrates the two critical dimensions of content emotions and comment semantics to discern news veracity. Its predictions are dynamically influenced by the interaction between questioning frequencies and the emotional context of the news.



(a) Semantic distribution of comments for non-sadness news.



(b) Semantic distribution of comments for sadness news.

Fig. 9. Semantic distribution of comments.**Fig. 10.** Prediction comparison by emotion and questioning frequency.

In summary, the results demonstrate that ESIN has successfully learned to identify relationships between the semantic distribution of comments and news emotions. This ability enables the model to provide more effective predictions grounded in the interplay between content emotion and comment semantics from the perspective of questions and non-questions.

4.7. Discussion of results

The above experimental results collectively validate that the proposed ESIN model effectively addresses the core research objective: to explore the latent associations between news content emotion and comment semantics, with a specific focus on distinguishing between questioning and non-questioning comment semantics, and to leverage these associations for enhanced fake news detection.

The primary research goal emphasizes uncovering the interplay between content emotion and comment semantics (questioning vs. non-questioning) as a discriminative cue for fake news. The model comparison results directly demonstrate that ESIN outperforms sixteen competitive baselines, including emotion-unaware models (e.g., BiGRU, InfOp) and emotion-aware variants (e.g., BiGRU-EMO, CEUS), by significant margins. These gains stem from ESIN's unique ability to model the latent emotion-semantic interaction:

unlike baselines that treat emotion and semantics as isolated features, ESIN explicitly captures how content emotions shape the distribution of questioning/non-questioning comments, exactly the association highlighted in the research objective. The ablation study further reinforces the alignment with the latent emotion-semantic association objective. Removing key components of ESIN that enable emotion-semantic interaction (e.g., Content Emotion, Semantic Quantification) leads to substantial performance declines. These results confirm that the interaction between content emotion and comment semantic distribution is a core mechanism that drives ESIN's effectiveness, directly addressing the research objective of leveraging such latent associations. A critical dimension of the research objective is distinguishing between questioning and non-questioning comment semantics and their differential associations with content emotion. The case study experiment further provides empirical evidence for this distinction. In addition, the core hyperparameters of the model are analyzed.

In summary, the experimental results collectively demonstrate that ESIN fully addresses the research objective. The model comparison and ablation studies confirm that the interaction between content emotion and comment semantics is a powerful discriminative cue; the case study and distribution visualization validate that questioning vs. non-questioning semantics exhibit emotion-dependent patterns that distinguish fake from real news. By explicitly modeling these latent interactions, ESIN not only achieves state-of-the-art performance but also advances the understanding of how user engagement (via comments) is shaped by news content emotion, a key insight highlighted in the research objective.

4.8. Theoretical and practical implications

4.8.1. Theoretical implications

A central challenge in fake news detection is the development of strategies that effectively extract and represent discriminative features from both the textual content of news and the user comments it generates. Prior research has highlighted that emotional cues embedded in news articles and their associated comments can serve as indicators of veracity. Nevertheless, existing approaches frequently model semantic and emotional signals in isolation, thereby neglecting the ways in which the emotion conveyed by a news item arouses the semantics of user comments, particularly the linguistic distinction between questioning and non-questioning forms of engagement. Our investigation demonstrates that fake news elicits comment semantics that diverge markedly from those associated with real news, with these divergences being conditioned by the emotional valence of the content. This interplay between content emotion and comment semantics constitutes a salient yet underexploited signal for telling real news apart from fake. Existing approaches, by not explicitly capturing this relationship, overlook a valuable source of discriminative insight, thereby limiting their effectiveness.

To bridge this gap, we introduce the ESIN, a framework designed to capture latent associations between news content emotion and comment semantics, with particular attention to the contrast between questioning and non-questioning comments. Methodologically, ESIN incorporates two novel components: Codebook Initialization, which enables the learning of structured semantic representations, and Semantic Quantification, which captures fine-grained distributions of comment semantics. These quantified semantic patterns are subsequently integrated with content emotions via a cross-attention mechanism, yielding a unified representation of how emotional cues shape the semantics of audience responses. This constitutes a departure from prevailing approaches that treat emotion and semantics separately. We conduct extensive experiments on two benchmark datasets to evaluate ESIN. The results reveal substantial improvements over competitive baselines, demonstrating the efficacy of explicitly modeling emotion-semantic interactions. In particular, ESIN excels in capturing the differential roles of questioning and non-questioning comments, a dimension overlooked in prior work. By systematically modeling this interaction, ESIN leverages the association between content emotion and comment semantics, thereby significantly enhancing the detection of fake news.

This finding fills an important theoretical gap by demonstrating that ESIN effectively addresses a key limitation of existing methods: the tendency to overlook how a news content's emotion interacts with comment semantics, particularly the distinction between questioning and non-questioning language. As a result, ESIN improves model performance in fake news detection. In addition, our finding aligns with the Affect and Cognition in Persuasion theory. We show fake news manipulates emotional cues to regulate cognitive engagement: non-sad emotions in fake content drive questioning comments, while sad fake news suppresses such scrutiny. This advances the theory by revealing how misinformation strategically uses emotion to shape cognitive responses. Our finding also aligns with the Uses and Gratifications theory. Fake news exploits user needs differently: non-sad fake news gratifies cognitive engagement (via questioning), while sad fake news caters to non-questioning responses. ESIN quantifies these patterns, operationalizing gratification-seeking in misinformation contexts.

4.8.2. Practical implications

Our work carries several important practical implications with direct relevance to real-world applications such as social media platforms and political discourse, where it can be integrated into monitoring systems to enhance fake news detection and strengthen media literacy.

First, these findings offer practical guidance for platform decision-makers. ESIN helps spot high-risk patterns that need closer attention. For example, fake news with non-sad emotions (e.g., anger-driven political fake news) should be prioritized when there is a surge in questioning comments, as this reflects public skepticism and may require quick action to prevent spread. On the other hand, fake news with sad content (e.g., false disaster stories) can be harder to detect because it often reduces questioning. In these cases, non-questioning comments, such as expressions of sympathy or support, should be monitored carefully, since they can unintentionally hide and normalize misinformation.

Second, the results yield implications for policymakers and regulators. By revealing how emotional cues shape comment semantics, ESIN underscores the importance of equipping users with the ability to recognize emotion–semantic dynamics. Regulators may further leverage ESIN’s semantic distribution analysis to establish evidence-based guidelines for platform accountability. For example, platforms could be required to ensure transparency regarding the prevalence of high-risk emotion–comment patterns, thereby operationalizing accountability mechanisms in content moderation.

Finally, by explicitly modeling the interplay between emotion and semantics, this study lays a methodological foundation for the next generation of fake news detection systems. Beyond improving detection accuracy, ESIN contributes to building resilience in the information ecosystem by reinforcing user awareness, promoting responsible platform governance, and ultimately strengthening trust in digital media environments.

5. Conclusion and future work

We introduce an Emotion-Semantic Interaction Network (ESIN), which takes into account the influence of questioning and non-questioning semantics in comments. It learns the latent relationship between content emotion and comment semantics from both questioning and non-questioning perspectives. Specifically, ESIN integrates the interaction between content emotions and the original comment semantics, along with the extracted distribution of semantic categories (questions and non-questions) in the comments. This key distribution is realized through two proposed mechanisms: Codebook Initialization and Semantic Quantification. To evaluate the effectiveness of ESIN, we conduct extensive experiments on two widely used fake news detection datasets, RumourEval and WeiBo, and the results demonstrate the method’s efficacy. While the proposed approach delivers promising performance, it is restricted to news articles with accompanying comments and relies exclusively on the text modality, missing opportunities to leverage other valuable sources of information, such as images. To overcome these limitations, our future work will focus on developing a versatile multi-expert framework. This framework will seamlessly integrate various fake news detection methods, including content-based and multi-modal approaches, enabling it to adapt to a broader range of application scenarios.

CRedit authorship contribution statement

Zhenhua Tan: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tao Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Data availability

Our original codes/data will be released at <https://github.com/TAN-OpenLab/ESIN>.

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